Conditional Probability and Bayes' Rule

50 minutes • 50 points • Student quiz

Name: _____ **Date:** _____

Instructions. Show your reasoning and give exact fractions. No calculator is needed. Show the Bayes' Rule setup in Problems 3(c) and 4(d). The suggested times leave 4 minutes to check your work.

1. Warm-up: a finite probability space

8 points · about 7 minutes

A bag contains eight tickets labeled a, b, b, c, c, d, d, d. Choose one ticket uniformly at random and record its letter. An event is any subset of the sample space of recorded letters.

(a) Give the sample space Ω of recorded letters and the probability of each outcome. Check that these probabilities sum to 1. (4 points)

(b) Let $A = \{a, c\}$ and $B = \{c, d\}$. Find $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cup B)$. (2 points)

(c) A student claims $\mathbb{P}(A) = 2/4$ because A contains two of the four possible letters. Is this correct? Give $\mathbb{P}(A)$ and explain. (2 points)

2. Which group is the denominator?

10 points · about 10 minutes

The table describes a group of 60 students. Choose one of them uniformly at random. Let C mean "belongs to the chess club" and M mean "plays a musical instrument."

	M	M^c	Total
C	12	18	30
C^c	8	22	30
Total	20	40	60

(a) Find $\mathbb{P}(C)$, $\mathbb{P}(M)$, and $\mathbb{P}(C \cap M)$. (3 points)

(b) Find $\mathbb{P}(M \mid C)$ and $\mathbb{P}(C \mid M)$. For each calculation, say which group supplies the denominator. (4 points)

(c) Are C and M independent? Justify your answer using probabilities. (3 points)

3. A red ball: which box?

12 points · about 11 minutes

Box U contains three red balls and one blue ball. Box V contains one red ball and three blue balls. Choose box U with probability $1/3$ and box V with probability $2/3$. Then draw one ball uniformly at random from the chosen box. Let U and V also denote the events that the corresponding box is chosen, and let R denote the event that the ball is red.

(a) List the four possible outcomes of the form (chosen box, ball color) and find their probabilities. Show how the probabilities for box choice and color combine. A labeled tree or table is fine. (4 points)

(b) Use the law of total probability to find $\mathbb{P}(R)$. (3 points)

(c) The drawn ball is red. Use Bayes' Rule to find the probability that box U was chosen. (3 points)

(d) Another student answers part (c) with $3/4$, saying "three of the four balls in box U are red." Which conditional probability does $3/4$ describe, and how does it differ from the probability requested? (2 points)

4. Monty Hall, step by step

20 points · about 18 minutes

A car is placed behind one of three doors, with probability 1/3 for each door. The other two doors hide goats. You initially choose Door 1. The host follows these rules:

- The host knows where the car is and always opens exactly one unchosen door that hides a goat.
- If both unchosen doors hide goats, the host chooses between them with equal probability 1/2.
- The host always offers you the choice of staying with Door 1 or switching to the other unopened door.

You observe the host open Door 3, revealing a goat. Let C_i mean "the car is behind Door i," and let H_3 mean "the host opens Door 3." All probabilities below assume your initial choice is Door 1.

(a) Before the host opens a door, what are $\mathbb{P}(C_1)$, $\mathbb{P}(C_2)$, and $\mathbb{P}(C_3)$? (2 points)

(b) Fill in the table and briefly explain each entry using the host's rules. These are probabilities of the host's action *given the car's location*. (6 points)

Car location	C_1	C_2	C_3
$\mathbb{P}(H_3 \mid C_i)$			

(c) Use the law of total probability to find $\mathbb{P}(H_3)$. Show all three car-location terms. (4 points)

(d) Use Bayes' Rule to find $\mathbb{P}(C_1 \mid H_3)$ and $\mathbb{P}(C_2 \mid H_3)$. Also state $\mathbb{P}(C_3 \mid H_3)$. (5 points)

(e) Should you stay with Door 1 or switch to Door 2? Give the chance of winning under each choice. Explain why "two doors remain, so each has probability 1/2" is not a valid argument. (3 points)